

The allegory axioms, and what defines them in the Frobenius calculus

§1. Cartesian bicategory of relations

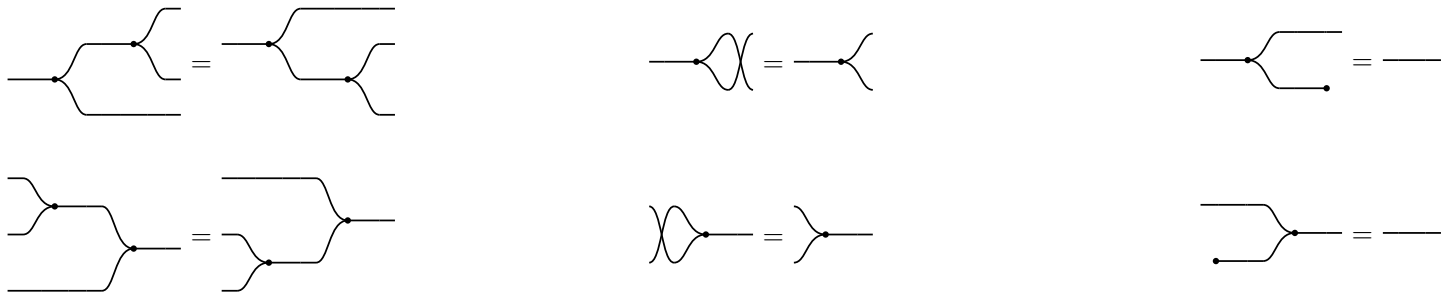
Definition 1.1.

A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product $\otimes$ (cartisn product of set in Rel, vertical junctposition in the diagram) and

$\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$, **Frobenius**, arrows **lax**.

$(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$ formed a cocommutative comonoid

$(\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ formed a commutative monoid



In $\text{Rel}(\text{Set})$, the four generators are these relations on a set A :

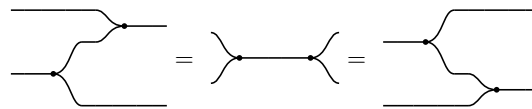
- $\triangleleft = \{(a, (a, a))\}$ — copy: one in, two identical out
- $\triangleright = \{((a, a), a)\}$ — merge: a pair passes only when its two components agree, and the common value comes out
- $\multimap = A \times \{*\}$ — discard: forget the value, keep only that there was one
- $\multimap = \{*\} \times A$ — create: any element at all

§1.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$



The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In Rel it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§1.2. The Frobenius law



The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, $\triangleright \triangleleft$ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, $\triangleleft \triangleright$: copy then merge, a closed loop, which the idempotency of n uses below.

§1.3. Every arrow is a lax comonoid homomorphism

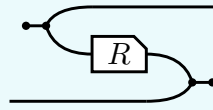


$$R\triangleleft \leq \triangleleft(R\otimes R)$$

§2. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

Definition 2.1.

The **converse** R° of $R : a \rightarrow b$ is R with both of its wires turned round,

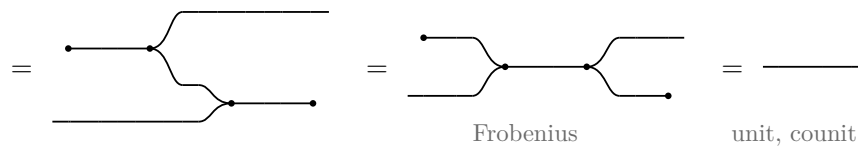


$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. And it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

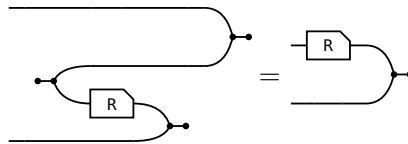
$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§2.1. $1^\circ = 1$ (snake)



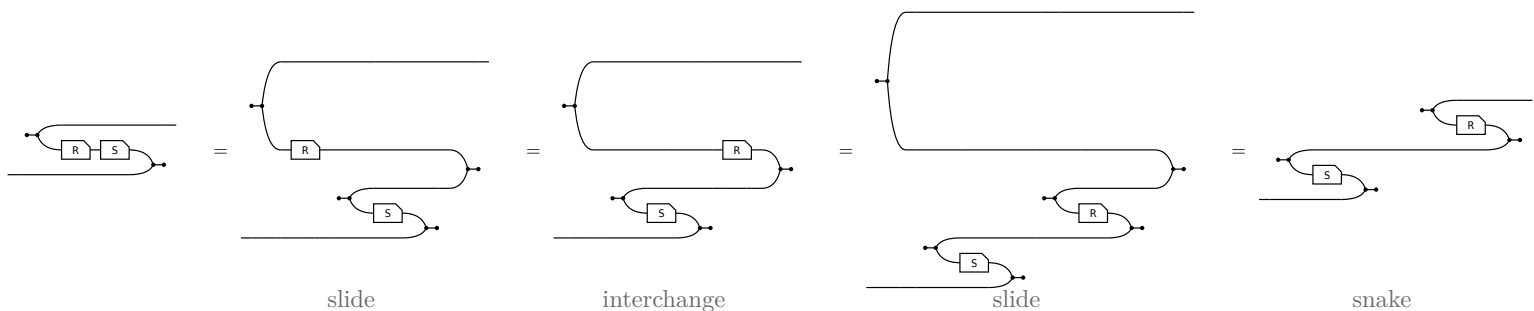
§2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



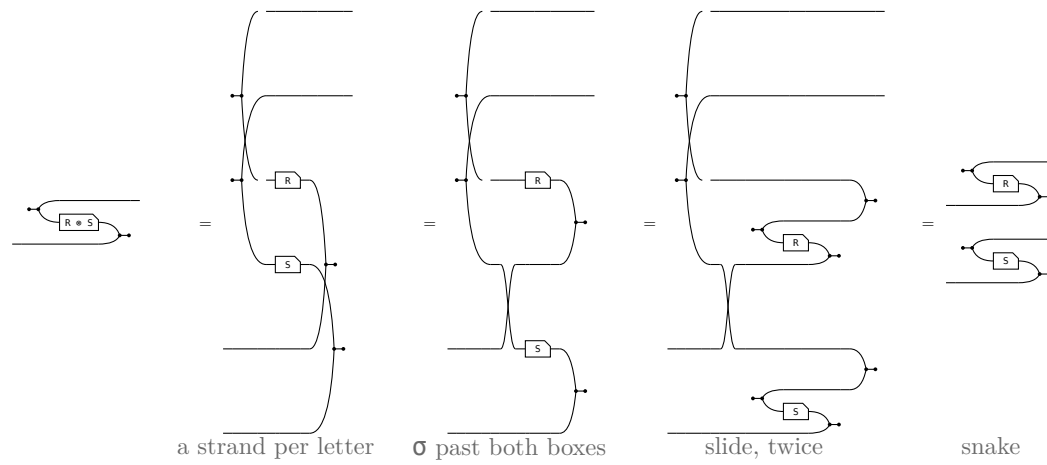
R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§2.3. $(R \ S)^\circ = S^\circ R^\circ$



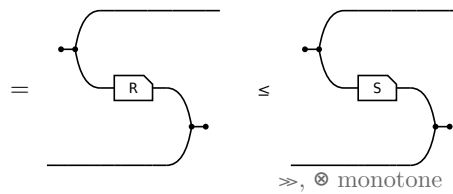
§2.4. $(R \otimes S)^\circ = R^\circ \otimes S^\circ$

A merge at a product is two merges behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.



§2.5. $R \leq S$ implies $R^\circ \leq S^\circ$

R sits in a frame of wires built from $\gg$ and $\otimes$, and both of those are monotone, so the box may be replaced where it stands.



§3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit $1 \leq \triangleleft \triangleright$, duplicate the box with the comonoid law, contract by the counit $\triangleright \triangleleft \leq 1$. Both halves of the adjunction, and nothing else.

In $\mathbf{Rel}(\mathbf{Set})$ with $R \subseteq A \times B$, where $\triangleright$ is the equality test on a pair and $\bullet$ produces any element at all:

$$\begin{aligned}
 \triangleright R &= \{((a, a), b) : a R b\} \text{ — two equal inputs, then one run of } R \\
 (R \otimes R) \triangleright &= \{((a, a'), b) : a R b \text{ and } a' R b\} \text{ — a run on each input, then the results merged} \\
 \bullet R &= \{(*, b) : \exists a. a R b\} \text{ — the image of } R \\
 \bullet &= \{*\} \times B \text{ — all of } B
 \end{aligned}$$

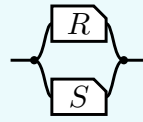
Take $a' = a$ for the first containment; every b lies in B for the second. Each is strict exactly when its reverse fails — R not injective, R not surjective — which is what the later table on maps reads off them.

inequation, and what proves it	picture
$(41) \triangleright R \leq (R \otimes R) \triangleright$ $\triangleright R = \triangleright R 1 \leq \triangleright R \triangleleft \triangleright \leq \triangleright \triangleleft (R \otimes R) \triangleright \leq (R \otimes R) \triangleright$ — unit, lax $\triangleleft$, counit.	
$(42) \bullet R \leq \bullet$ $\bullet R = \bullet R 1 \leq \bullet R \bullet \leq \bullet \bullet \leq \bullet$ — the same three steps at $\mathbb{I}$.	

§4. $\cap$ is a commutative idempotent monoid on every hom-set

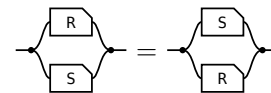
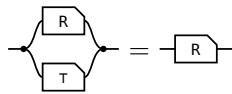
Definition 4.1.

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



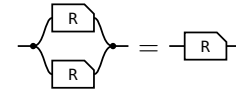
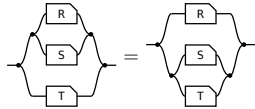
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$

commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Unitary and pre-tabular — what the other direction needs

condition	allegory	Frobenius side
unitary	τ is a unit when 1τ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes τ exist.	free. 1 is the unit object, $1 \leq \multimap \multimap$ says $\multimap$ is entire, and $\tau = \multimap \multimap$ is maximal.
pre-tabular	f, g tabulate R when both are maps, $R = f \circ g$, and R is tabular when some pair tabulates it, the allegory pre-tabular when every arrow lies under a tabular one.	this is what $\otimes$ is. Given unitarity it reduces to tabulating each τ , and a tabulation of τ is an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

Together: **cartesian bicategory of relations** $\simeq$ **unitary pre-tabular allegory**.

§6. Union

Past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure. It draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. The residual, which needs a second composition with linear adjoints, is in the division section below.

statement	picture
$R \sqsubseteq R \cup S$ union $R S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which $\cap$ does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	

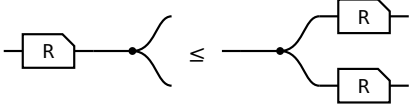
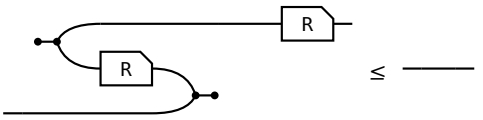
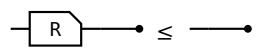
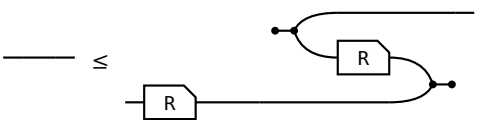
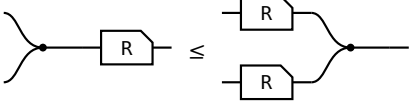
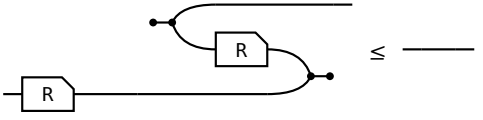
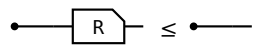
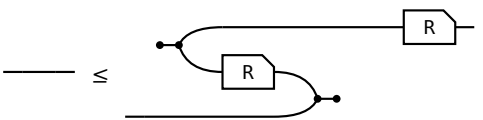
The last of Freyd's §2.21 equations that have no picture here — a tape around nothing, or a tape around what is already inside it, is not worth drawing. R = shops with milk, S = shops with bread throughout.

equation	the milk reading
$R \cup (S \cap R) = R \quad (R \cup S) \cap R = R$ absorption	Shops with milk, or with both: still the shops with milk. Shops with milk or bread, and with milk: the same. The smaller side of each pair is already inside the larger.

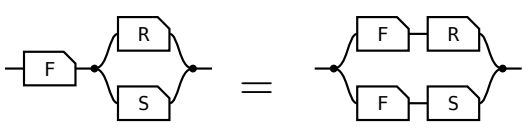
equation	the milk reading
$R \perp = \perp R$ $0_S = 0_{\{R S\}}$ — and $\perp R = \perp$ on the other side	Going to a shop and then taking a road that leads nowhere gets you nowhere. $\perp$ is a two-sided zero for composition, as τ is not.

§7. Maps

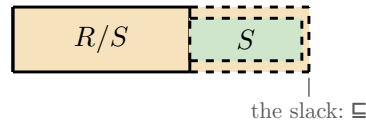
Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by the lax monoid section. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

holds for every arrow	property	holds iff
$R \triangleleft \leq \triangleleft (R \otimes R)$ 	(SV) single valued $R^\circ R \sqsubseteq 1$	
$R \multimap \leq \multimap$ 	(TOT) total $1 \sqsubseteq R^\circ R$. With single valued, a map.	
$\triangleright R \leq (R \otimes R) \triangleright$ 	(INJ) injective $R R^\circ \sqsubseteq 1$	
$\multimap R \leq \multimap$ 	(SUR) surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	

What (SV) buys, in one law. Only single valuedness is spent: F may be partial, and entirety is never used.

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = shops Ann may go to, R = has milk, S = has bread. Single valued means one shop, so the sides agree.	 one shop with both milk at A, bread at B

§8. Division



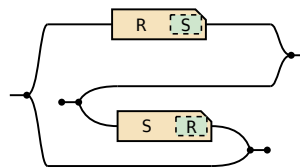
$T \sqsubseteq R/S \iff T S \sqsubseteq R$	
$T \sqsubseteq S \setminus R \iff S T \sqsubseteq R$	
$(R/S) S \sqsubseteq R$ Strict at $S = \emptyset$: R/S is everything, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$	
$(S T) \setminus R = T \setminus (S \setminus R)$	
maps: $f (R/S) = (f R)/S$ The same three pieces bracketed two ways, which is the licence to write $f R / S$. The shunting rule.	
$R/(f S) = (R/S) f^\circ$ A map moves out of a denominator and reappears as f° outside the box. The shunting rule again.	
§2.314: $(R/S) (S/T) \sqsubseteq R/T$	
$1 \sqsubseteq R/R$ R = shop a stocks c , so R/R = “ a stocks all a ’ does”, true of every shop. Strict: two shops stocking only milk contain each other and stay two shops.	
$(R/R) (R/R) = R/R$ Freyd writes $\sqsubseteq$, but with $1 \sqsubseteq R/R$ above it is an equality: $R/R = 1 (R/R) \sqsubseteq (R/R) (R/R)$ is the other direction. A preorder is idempotent.	
$(R/R) R = R$	
$R/1 = R$	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ A denominator’s union becomes a numerator’s meet: dividing by more leaves less.	
$S \setminus (R/T) = (S \setminus R)/T$ Which is why $S \setminus R/T$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. Symmetric division

$R/_s S \triangleq (R/S) \cap (S/R)^\circ$, the two-sided quotient: everything R says about a column and everything S says about it, in both directions. In $\mathbf{Rel}$ it relates x to y when the two are related to the **same** things, $\forall z. (x R z \iff y S z)$ — the shops again: x stocks under R exactly what y stocks under S .

Freyd writes it as a fraction, R over S : $\frac{R}{S}$. The definition drawn is a meet of two long divisions, the second turned round by the converse frame:



$\triangleleft ((R/S) \otimes (S/R)^\circ) \triangleright$ — exported from the definition, not transcribed

$(R/_s S)^\circ = S/_s R$ The meet is inside the definition, not the statement, so it stays under the label.	
$(R/_s S)(S/_s T) \sqsubseteq R/_s T$ $/_s$ is converse-symmetric and transitive, like an equality of columns.	

The rest of §2.35, without pictures: $/_s$ is not built from the generators, so a dashed box says no more than the formula does. Equations are written as equations even where Freyd writes $\sqsubseteq$.

law	what it says
$T \sqsubseteq R/_s S \iff T S \sqsubseteq R \text{ and } T^\circ R \sqsubseteq S$	The characterization — one condition per half of the meet. $/_s$ is partial : both halves must typecheck.
$(R/_s S) S \sqsubseteq R \quad (R/_s R) R \sqsubseteq R$	Cancellation, one bar's slack. Shops with the same stock, fed back through S , reach no more than R did.
$1 \sqsubseteq R/_s R \quad (R/_s R)(R/_s R) = R/_s R$	$R/_s R$ is reflexive and transitive , and symmetric by the row above, so it is an equivalence relation : same stock. Freyd writes $\sqsubseteq$ for the square; with $1 \sqsubseteq R/_s R$ it is an equality, exactly as for R/R .
$T \sqsubseteq R/_s R \iff T R \sqsubseteq R$, for symmetric T (§2.351)	What picks $R/_s R$ out among the symmetric arrows: the largest one R is invariant under.
$S/_s S = 1$, S is straight	No two things have the same S -stock. Equivalently every symmetric T with $T S \sqsubseteq S$ is coreflexive.
$f S = g S \implies f = g$, S straight (§2.352)	A straight S cancels on the right. A converse holds where every morphism is a union of semisimple ones, §2.353.
$S R \text{ straight} \implies S \text{ straight}$ (§2.355)	Straightness passes to a prefix; in particular a right-invertible S is straight.
$S \text{ straight} \iff R/S \text{ simple for all } R$ (§2.356)	$(R/S)^\circ(R/S) \sqsubseteq (S/_s R)(R/_s S) \sqsubseteq S/_s S \sqsubseteq 1$.
$R = h S$, h a cover, S straight (§2.354)	In an effective division allegory every arrow factors that way.
$R/_s 1$ is the simple part of R (§2.357)	$x (R/_s 1) y$ iff x 's image is exactly $\{y\}$ — the shops stocking one item and nothing else. So $R/_s 1 = R$ iff R is simple, unlike $R/1 = R$, which is unconditional.

law	what it says
$\text{Dom } (R/_s S) = 1 \cap (R/S)(S/R)$ (§2.357)	Its domain is the domain of simplicity of R.